LITERATURE SURVEY

NEXT-TOKEN PROBABILITY EXPLORER FOR MODEL UNCERTAINTY ANALYSIS

Generative Artificial Intelligence (GenAI)

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1. Introduction

Generative Artificial Intelligence (GenAI) and Large Language Models (LLMs) have improved the ability of machines to understand and generate human-like text. Many widely used language models generate text autoregressively: at each generation step, a model estimates a probability distribution over possible next tokens, conditioned on the tokens that came before.

The next-token distribution contains useful information about the model's immediate prediction. If one token has a much higher probability than the alternatives, the distribution is concentrated. If probability is spread across many alternatives, the prediction is more uncertain. Measures such as maximum token probability and Shannon entropy can help describe this distribution.

This literature survey reviews three research papers related to the Transformer architecture, the mechanics of next-token prediction, and uncertainty quantification for large language models. Together they provide the theoretical background for the proposed Next-Token Probability Explorer for Model Uncertainty Analysis: the first explains how a model produces next-token probabilities, the second explains what a self-attention model learns when trained to predict the next token, and the third explains how token probabilities and entropy are used to measure uncertainty, and where they fall short.

2. Literature Review

2.1 Attention Is All You Need (2017)

Authors: Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Łukasz Kaiser, and Illia Polosukhin | Venue: 31st Conference on Neural Information Processing Systems (NIPS/NeurIPS), 2017.

Literature review: Vaswani et al. propose the Transformer, a sequence transduction architecture based solely on attention, dispensing with recurrence and convolution. Both the encoder and the decoder are stacks of N = 6 identical layers built from multi-head self-attention and position-wise feed-forward sub-layers, with residual connections and layer normalization. Attention is computed as softmax($QK^T/\sqrt{d_k}$)V (scaled dot-product attention), run in parallel over h = 8 heads, and sinusoidal positional encodings supply word-order information. The decoder uses masked self-attention so that the prediction at position i depends only on earlier positions, preserving the auto-regressive property. The decoder output passes through a linear layer and a softmax to produce predicted next-token probabilities. On WMT 2014 the big model reached 28.4 BLEU (English–German) and 41.8 BLEU (English–French) after 3.5 days of training on eight GPUs, at a fraction of the training cost of earlier models, and it also generalized to English constituency parsing. The authors additionally note that label smoothing ($\varepsilon = 0.1$) hurts perplexity because the model learns to be more unsure, while improving accuracy and BLEU.

Relevance to the proposed project: This paper explains how context is processed and how the final softmax layer turns the decoder output into a probability distribution over the vocabulary, which is

exactly the distribution the explorer will display. Its remark that label smoothing makes a model more unsure is a useful reminder that output probabilities depend on training choices and are not objective measures of certainty. It is a foundational reference rather than a study of uncertainty or visualization.

Source: https://arxiv.org/abs/1706.03762

2.2 Mechanics of Next Token Prediction with Self-Attention (2024)

Authors: Yingcong Li, Yixiao Huang, M. Emrullah Ildiz, Ankit Singh Rawat, and Samet Oymak | Venue: Proceedings of the 27th International Conference on Artificial Intelligence and Statistics (AISTATS), PMLR vol. 238, 2024.

Literature review: This theoretical paper asks what a single self-attention layer learns from next-token prediction. Studying a one-layer self-attention model with a linear classification head trained by gradient descent, the authors show that it learns an automaton that generates the next token in two steps: hard retrieval, where attention selects the high-priority tokens associated with the last input token, and soft composition, where it forms a convex combination of those tokens from which the next token can be sampled. Priority is formalized through token-priority graphs (TPGs) built from the training data, and the model is shown to discover their strongly-connected components (SCCs). The authors prove global directional convergence of gradient descent under log-loss to the solution of a support-vector-machine formulation (Graph-SVM), characterize the implicit bias of the regularization path, and show that convergence can be only local under more general settings. Synthetic experiments (for example K = 6, d = 8) confirm the theory, with correlation between the learned weights and the SVM solution reaching about 0.987.

Relevance to the proposed project: The paper provides a principled account of why a model assigns high probability to some candidate tokens and suppresses others, and of how probability is shared among tokens of equal priority. This helps in interpreting the distributions the explorer will show. However, its analysis is limited to a single attention layer under stated assumptions, so it supports the project's understanding of next-token prediction rather than describing the behaviour of full-scale language models or offering an uncertainty measure.

Source: Proceedings of Machine Learning Research (PMLR), vol. 238, AISTATS 2024 (provided PDF: Research_paper_2)

2.3 A Survey on Uncertainty Quantification of Large Language Models: Taxonomy, Open Research Challenges, and Future Directions (2025)

Authors: Ola Shorinwa, Zhiting Mei, Justin Lidard, Allen Z. Ren, and Anirudha Majumdar | Publication: ACM, 2025 (arXiv:2412.05563).

Literature review: This survey reviews uncertainty quantification (UQ) methods for LLMs and organizes them into four categories: token-level UQ, self-verbalized UQ, semantic-similarity UQ, and mechanistic interpretability. It distinguishes aleatoric from epistemic uncertainty and separates white-box methods, which use the model's token probabilities, from black-box methods, which use only generated text. The white-box metrics it lists include average token log-probability, perplexity,

maximum token log-probability, response improbability, and entropy of the token probability distribution, $H = -\Sigma\, p(w) \log p(w)$. It notes that token-based scores degrade for long responses and discusses length-normalized alternatives. It also covers applications such as hallucination detection and robotics, and identifies open challenges. Importantly, the authors argue that "entropy is not factuality" and "consistency is not factuality": a model can assign high token probability to an incorrect answer (for example, 91% token-level confidence for a false claim about the most populous country), and RLHF fine-tuning generally reduces the calibration of confidence estimates.

Relevance to the proposed project: This is the key reference for the project's uncertainty component. It directly supports exposing candidate-token probabilities, the maximum probability, and Shannon entropy as uncertainty indicators, and it justifies the project's requirement to state clearly that token confidence is not a guarantee of factual correctness. Its taxonomy also identifies semantic-similarity and self-verbalized methods as possible future extensions beyond token-level analysis.

Source: https://arxiv.org/abs/2412.05563

3. Comparative Analysis of Existing Research

Research paper	Main contribution	Use in proposed project
Vaswani et al. (2017)	Transformer and self-attention architecture; softmax output over next-token probabilities	Explain contextual processing and how the probability distribution is produced
Li et al. (2024)	Theory of what single-layer self-attention learns in next-token prediction (hard retrieval, soft composition, SVM bias)	Support understanding of why next-token probabilities take the shapes they do
Shorinwa et al. (2025)	Taxonomy of LLM uncertainty quantification, token-level metrics such as entropy, and their limits	Guide entropy and confidence indicators and the caveat that entropy is not factuality

4. Existing System

The reviewed papers address different aspects of the problem: the Transformer paper defines the architecture that produces next-token probabilities, the self-attention paper analyses what such a model learns when trained for next-token prediction, and the uncertainty-quantification survey catalogues methods (including token probability and entropy) for measuring a model's uncertainty. These works provide important technical foundations, but they are theoretical or survey-oriented and are not designed as a single, beginner-friendly interactive exploration tool.

5. Research Gap

The reviewed literature establishes the importance of next-token prediction and of uncertainty estimation, and the survey in particular shows that token-level metrics such as entropy are widely used but must be interpreted with care. However, there is an opportunity to bring several related ideas together in an accessible interactive platform. The proposed project aims to combine candidate-token probability display, maximum-probability indicators, entropy calculation, uncertainty categorization, and graphical comparison of prompts in one interface. This is the proposed project's design objective; it should not be interpreted as a claim that no similar tools or systems exist.

6. Proposed System

The Next-Token Probability Explorer is proposed as an interactive application that accepts a text prompt, passes it to a language model capable of returning next-token scores, and displays the model's candidate next tokens with their probabilities. The application calculates entropy from the probability distribution and presents the results using charts and concise explanations.

1. Prompt input: accept text entered by the user.
2. Tokenization and model processing: prepare the prompt and obtain model outputs.
3. Probability extraction: retrieve candidate next-token scores or probabilities from the model.
4. Probability normalization: convert scores such as logits into a probability distribution using softmax, where required.
5. Entropy calculation: calculate Shannon entropy from the token probabilities.
6. Uncertainty display: show a clearly defined low, medium, or high uncertainty category using documented thresholds.
7. Visualization: display candidate tokens, probability percentages, and comparisons across prompts.

7. Objectives

- To understand the next-token prediction mechanism used by autoregressive language models.
- To obtain and display probability distributions for candidate next tokens.
- To calculate uncertainty using Shannon entropy.
- To display the highest-probability next token and its probability.
- To compare probability distributions and entropy values for different prompts.
- To explain the limitations of token probability as a measure of factual correctness.

8. Methodology

8. **User input:** The user enters a prompt into the application.

9. **Model inference:** The prompt is passed to a suitable pretrained language model that exposes next-token scores or probabilities.
10. **Probability processing:** If the model returns logits, apply softmax to obtain normalized probabilities.
11. **Entropy calculation:** Calculate Shannon entropy: $H = -\Sigma\, p_i \log_2(p_i)$, where p_i is the probability of candidate token i.
12. **Confidence indicator:** Display the maximum candidate-token probability as a simple next-token confidence indicator.
13. **Uncertainty classification:** Map entropy or normalized entropy to low, medium, or high categories using thresholds selected and explained by the project team.
14. **Visualization and comparison:** Show a ranked token list or bar chart and compare results for multiple prompts.

9. Expected Results

The expected result is a web-based interactive application that displays the next-token probability distribution produced by a selected language model. Users will be able to inspect likely candidate tokens, view their probability values, observe entropy-based uncertainty indicators, and compare how different prompts affect the distribution. The application is intended as an educational analysis tool; its token-level confidence indicator will not be presented as a guarantee of factual accuracy.

10. Limitations and Considerations

- Not every hosted language-model API exposes the full next-token probability distribution; model and API selection must account for this.
- Token probabilities depend on the model, tokenizer, prompt, decoding settings, and any applied temperature or sampling configuration.
- Maximum next-token probability describes the model's immediate token distribution, not the truth of the complete response.
- Entropy depends on the candidate set and probability distribution. Comparisons should use consistent settings and explain the chosen normalization.
- Low, medium, and high uncertainty thresholds are design choices and should be justified or calibrated using experiments.

11. References / Bibliography

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